

USER INFORMATION
DOPAINE
STERILE DOPAMINE CONCENTRATE 40 mg/mL
Sterile and Non-Pyrogenic

Composition:

Each mL solution contains 40 mg Dopamine Hydrochloride corresponding to 200 mg in 5 mL.

Product Description:

DOPAINE is a colourless or pale-yellow solution.

Pharmacology:

Pharmacodynamics:

Dopamine stimulates adrenergic receptors of the sympathetic nervous system. The drug has principally a direct stimulatory effect on β_1 -adrenergic receptors, but also appears to have an indirect effect by releasing Norepinephrine from its storage sites. Dopamine also act on specific dopaminergic receptors in the renal, mesenteric, coronary, and intracerebral vascular beds to cause vasodilation. The drug has little or no effect on β_2 -adrenergic receptors.

Pharmacodynamic effects:

In IV doses of 0.5 - 2 microgram/kg per minute, the drug acts predominantly on dopaminergic receptors; in IV doses of 2 - 10 microgram/kg per minute, the drug also stimulates β_1 -adrenergic receptors. In higher therapeutic doses, α -adrenergic receptors are stimulated and the net effect of the drug is the result of α -adrenergic, β_1 -adrenergic, and dopaminergic stimulation. The main effects of Dopamine depend on the dose administered. In low doses, cardiac stimulation and renal vascular dilation occur and in larger doses vasoconstriction occurs. It is believed that α -adrenergic effects result from inhibition of the production of cyclic adenosine-31, 51-monophosphate (cAMP) by inhibition of the enzyme adenylyl cyclase, whereas β -adrenergic effects result from stimulation of adenylyl cyclase activity.

Pharmacokinetics:

Absorption:

Orally administered Dopamine is rapidly metabolised in the gastrointestinal tract. Following IV administration, the onset of action of Dopamine occurs within 5 minutes and the drug has duration of action of less than 10 minutes.

Distribution:

The drug is widely distributed in the body but does not cross the blood-brain barrier to a substantial extent. It is not known if Dopamine crosses the placenta.

Elimination:

Dopamine has a plasma half-life of about 2 minutes. Dopamine is metabolised in the liver, kidneys, and plasma by monoamine oxidase (MAO) and catechol-O-methyltransferase to the inactive compounds homovanillic acid (HVA) and 3, 4-dihydroxyphenylacetic acid. In patients receiving MAO inhibitors, the duration of action of Dopamine may be as long as 1 hour. About 25% of a dose of Dopamine is metabolised to norepinephrine within the adrenergic nerve terminals.

Dopamine is excreted in urine principally as HVA and its sulfate and glucuronide conjugates and as 3, 4-dihydroxyphenylacetic acid. A very small fraction of a dose is excreted unchanged. Following administration of radio labelled Dopamine, approximately 80% of the radioactivity reportedly is excreted in urine within 24 hours.

Indication:

Dopamine Hydrochloride is indicated for the correction of haemodynamic imbalances present in the shock syndrome due to myocardial infarctions, trauma, endotoxic septicaemia, open heart surgery, renal failure and chronic cardiac decompensation as in congestive failure. Poor perfusion of vital organs. Urine flow appears to be one of the better diagnostic signs by which adequacy of vital organ perfusion can be monitored. Nevertheless, the physician should also observe the patient for signs of reversal of confusion or comatose condition. In a number of oliguric or anuric patients, administration of Dopamine Hydrochloride has resulted in an increase in urine flow which in some cases reached normal levels.

Dopamine Hydrochloride may also increase urine flow in patients whose output is within normal limits and thus maybe of value in reducing the degree of pre-existing fluid accumulation. It should be noted that at doses above those optimal for the individual patient, urine flow may decrease, necessitating reduction of dosage. Concurrent administration of Dopamine Hydrochloride and diuretic agents may produce an additive or potentiating effect. Low cardiac output. Increased cardiac output is related to the direct inotropic effect on the myocardium by Dopamine Hydrochloride. Increased cardiac output at low or moderate doses appears to be related to a favourable prognosis. In many instances, the renal fraction of the total cardiac output has been found to increase. Increase in cardiac output produced by Dopamine Hydrochloride is not associated with substantial decreases in systemic vascular resistance (SVR) as may occur with isoprenaline. Hypotension due to inadequate cardiac output can be managed by administration of low to moderate doses of Dopamine Hydrochloride which have little effect on SVR. At high therapeutic doses, the alpha adrenergic activity of Dopamine Hydrochloride becomes more prominent and thus may correct hypotension due to diminished SVR. It is suggested that the physician administer Dopamine Hydrochloride as soon as a definite trend toward decreased systolic and diastolic pressure becomes evident.

Recommended Dosage:

ADULT

IV Initial: 2.5 mcg/kg/min in patients who are likely to respond to modest increments of heart force and renal perfusion. In more severe cases, administration may begin with 5 mcg/kg/min and increased gradually in 5-10 mcg/kg/min increments up to 20 to 50 mcg/kg/min as needed.

It is recommended to check urine output frequently, if doses in excess of 50 mcg/kg/min are necessary.

In patients who do not respond to these doses, additional increments of dopamine may be given in an effort to achieve adequate blood pressure, urine flow and perfusion generally.

CHILDREN

The safety and efficacy of dopamine in paediatric patients has not been established.

GERIATRIC

No difference in dosage is suggested for geriatric patients. However, close monitoring is suggested for blood pressure, urine flow and peripheral tissue perfusion.

DOPAINE is a concentrated solution and MUST BE DILUTED BEFORE USED with the appropriate diluents;

- Sodium Chloride 0.9% w/v Intravenous Infusion
- Dextrose 5% w/v Intravenous Infusion
- Sodium Chloride 0.9% w/v and Dextrose 5% w/v Intravenous Infusion
- Sodium Chloride 0.45% w/v and Dextrose 5% w/v Intravenous Infusion
- Compound Sodium Lactate (Hartmann's Solution) Intravenous Infusion

Route of administration: Intravenous (IV)

Contraindications:

DOPAINE is contraindicated in:

- hypersensitivity to the active substance(s) or to any of the excipients in this product.
- patients with phaeochromocytoma or hyperthyroidism.
- the presence of uncorrected atrial or ventricular tachyarrhythmias or ventricular fibrillation.
- use with cyclopropane and halogenated hydrocarbon anaesthetics.

Warning and Precautions:

This preparation contains Sodium Metabisulphite that may cause serious allergic type reactions in certain susceptible patients. Do not use if known to be hypersensitive to Bisulphites

Dopamine should not be used in the presence of uncorrected tachyarrhythmias or ventricular fibrillation. Nor should it be used in patients with phaeochromocytoma or hyperthyroidism. Cyclopropane and halogenated hydrocarbon anaesthetics must be avoided.

Monoamine Oxidase (MAO) Inhibitors

Patients who have been treated with MAO inhibitors prior to Dopamine should be given reduced doses; the starting dose should be one tenth (1/10th) of the usual dose.

Potassium-Free Solutions

Excess administration of Potassium-free solutions may result in significant hypokalaemia. The intravenous administration of these solutions can cause fluid and/or solute overloading resulting in dilution of serum electrolyte concentrations, overhydration, congested states or pulmonary oedema.

Hypovolaemia

Hypovolaemia should be corrected where necessary prior to Dopamine infusion. Low doses should be used in shock due to acute myocardial infarction.

Decreased Pulse Pressure

If a disproportionate rise in diastolic pressure (i.e. a marked decrease in pulse pressure) is observed, the infusion rate should be decreased and the patients observed carefully for further evidence of predominant vasoconstriction activity, unless such an effect is desired.

Occlusive Vascular Disease

Patients with a history of peripheral vascular disease should be closely monitored for any changes in colour or temperature of the skin of the extremities. If change of skin colour or temperature occurs and is thought to be the result of compromised circulation to the extremities, the benefits of continued Dopamine Infusion should be weighed against the risk of possible necrosis. These changes may be reversed by decreasing the rate or discontinuing the infusion. IV administration of Phentolamine Mesylate 5-10 mg may reverse the ischaemia.

Extravasation

Dopamine Hydrochloride in 5% Dextrose Injection should be infused into a large vein whenever possible to prevent the possibility of infiltration of perivascular tissue adjacent to the infusion site. Extravasation may cause necrosis and sloughing of the surrounding tissue. Ischaemia can be reversed by infiltration of the affected area with 10-15 mL of Saline containing 5 to 10 mg Phentolamine Mesylate. A syringe with a fine hypodermic needle should be used to liberally infiltrate the ischaemic area as soon as extravasation is noted.

Diabetes

Dextrose solutions should be used with caution in patients with known subclinical or overt diabetes mellitus.

Renal and Hepatic Impairment

As the effect of dopamine on impaired renal and hepatic function is not known, close monitoring is advised.

Hypotension

Dopamine infusion should be withdrawn gradually to avoid unnecessary hypotension.

Laboratory Test Interferences

Infusion of Dopamine suppresses pituitary secretion of thyroid stimulating hormone, and prolactin. Dopamine should not be added to alkaline diluents.

Paediatric Use

The safety and efficacy of Dopamine in paediatric patients has not been established.

Excipient Information

This medicine contains Sodium Metabisulfite (E 223) which may rarely cause severe hypersensitivity reactions and bronchospasm. This medicine contains less than 1 mmol Sodium (23 mg) per vial, that is to say essentially 'sodium-free'.

Precautions:

DOPAMINE must be used immediately after dilution. Do not dilute with alkaline solutions. Single dose container. Discard all unused contents. Do not use if the solution is discolored and leakage is detected. If any visible solid particles, growth or turbidity appears during storage, the product should not be used.

Interaction with Other Medicaments

Anaesthetics

The myocardium is sensitised by the effect of Dopamine, cyclopropane or halogenated hydrocarbon anaesthetics, and these must be avoided. This interaction applies both to pressor activity and cardiac beta-adrenergic stimulation.

Alpha and Beta Blockers

The cardiac effects of Dopamine are antagonised by β -adrenergic blocking agents such as Propranolol and Metoprolol, and the peripheral vasoconstriction caused by high doses of dopamine is antagonised by α adrenergic blocking agents. Dopamine-induced renal and mesenteric vasodilation is not antagonised by either α or β -adrenergic blocking agents, but, in animals, is antagonised by haloperidol or other butyrophenones, phenothiazines, and opiates.

Monamine Oxidase (MAO) Inhibitors

MAO inhibitors potentiate the effect of dopamine and its duration of action. Patients who have been treated with MAO inhibitors prior to administration of Dopamine will therefore require a substantially reduced dosage. (The starting dose should be reduced to at least $1/10^{\text{th}}$ of the usual dose).

Phenytoin

Administration of IV Phenytoin to patients receiving Dopamine has resulted in hypotension and bradycardia; some clinicians recommend that Phenytoin be used with extreme caution, if at all, in patients receiving Dopamine.

Diuretic Agents

Dopamine may increase the effect of diuretic agents.

Ergot Alkaloids

The ergot alkaloids should be avoided because of the possibility of excessive vasoconstriction.

Tricyclic Antidepressants and Guanethidine

Tricyclic antidepressants and guanethidine may potentiate the pressor response to Dopamine.

Incompatibility:

DOPAMINE should not be added to any alkaline intravenous solutions, i.e. Sodium Bicarbonate. Any solution which exhibits physical or chemical incompatibility through a colour change or precipitate should not be administered.

It is suggested that admixtures containing Gentamicin Sulfate, Cephalothin Sodium, Cephalothin Sodium Neutral or Oxacillin Sodium should be avoided unless all other viable alternatives have been exhausted.

Admixtures of Ampicillin and Dopamine in 5% Dextrose Solution are alkaline and incompatible and result in decomposition of both drugs. They should not be admixed.

Admixtures of Dopamine, Amphotericin B in 5% Dextrose Solution are incompatible as a precipitate form immediately on mixing.

Pregnancy and Lactation:

Pregnancy

Animal studies have shown no evidence of teratogenic effects with Dopamine. However, the effect of Dopamine on the human foetus is unknown. Therefore, the drug should be used in pregnant women only when the expected benefits outweigh the potential risk to the foetus.

Lactation

It is not known if Dopamine is excreted in breast milk, nor is the effect on the infant known.

Side Effects:

Adverse reactions to Dopamine are related to its pharmacological action.

Frequencies are defined as: very common ($>1/10$), common ($>1/100$ to $<1/10$), uncommon ($>1/1,000$ to $<1/100$), rare ($>1/10,000$ to $<1/1,000$), very rare ($<1/10,000$), not known (cannot be estimated from the available data).

System Organ Class	Frequency	Adverse Reactions
Infections and infestations	Uncommon	Gangrene ¹
Nervous system disorders	Common	Headache
Eye disorders	Uncommon	Mydriasis
Cardiac disorders	Common	Ectopic heart beats, tachycardia, anginal pain, palpitation
	Uncommon	Aberrant conduction, bradycardia, widened QRS complex, fatal ventricular arrhythmias have been reported on rare occasions
	Not known	Atrial fibrillation
Vascular disorders	Common	Hypotension, vasoconstriction
	Uncommon	Hypertension
Respiratory, thoracic and mediastinal disorders	Common	Dyspnoea
Gastrointestinal disorders	Common	Nausea, vomiting
Skin and subcutaneous tissue disorders	Uncommon	Piloerection
Renal and urinary disorders	Uncommon	Azotaemia

¹Serious or Life-threatening Reactions: Gangrene of the feet has occurred following doses of 10-14 mcg/kg/min and higher in a few patients with pre-existing vascular disease.

Symptoms and Treatment of Overdose:

Excessive elevation of blood pressure and vasoconstriction can occur due to the alpha-adrenergic actions of Dopamine, especially in patients with a history of occlusive vascular disease. If desired, this condition can be rapidly reversed by dose reduction or discontinuing the infusion, since Dopamine has a half-life of less than two (2) minutes in the body. Should these measures fail, an infusion of an alpha-adrenergic blocking agent, e.g., Phentolamine Mesylate should be considered.

Dopamine at the infusion site can cause local vasoconstriction hence the desirability of infusing into a large vein. The resulting ischaemia can be reversed by infiltration of the affected area with 10 - 15 mL of Saline containing 5 mg to 10 mg Phentolamine Mesylate. A syringe with a fine hypodermic needle should be used to liberally infiltrate the ischaemic area as soon as extravasation is noted.

Storage Condition before and after reconstitution:

Do not store above 30°C. Protect from light.

Shelf Life:

2 years from manufacturing date in the proposed storage condition. The solution should be used within 24 hours after reconstitution. Do not use after expiry.

Dosage form and packaging available:

5 mL x 20 vials per box

Manufacturer / Product Registration Holder:



AIN MEDICARE SDN.BHD.

Lot 4933, 4934 & PT2464, Jalan 6/44, Kaw. Perindustrian Pengkalan Chepa 2, 16100 Kota Bharu, Kelantan, MALAYSIA.

Date of Revision: 25.06.2025